

MD HASANUR RAHMAN

PhD Candidate ◇ University of Florida ◇ mrahman7@ufl.edu ◇ hasanur-rahman.github.io

EDUCATION

University of Florida

2025–Present

PhD in Electrical and Computer Engineering

Continuing from Iowa

Advisor: Guanpeng Li

University of Iowa

2021–2025

PhD study in Computer Science

Got MS in Computer Science, 2024

Advisor: Guanpeng Li

Bangladesh University of Engineering and Technology

2015–2019

BS in Computer Science and Engineering

RESEARCH INTERESTS

High Performance Computing, Dependable Computing Systems, Data Reduction, Parallel Computing, Program Analysis, Large Language Model Inference

SELECTED RESEARCH CONTRIBUTIONS

- **Pessimistic Silent Error Evaluation for HPC Applications:** Developed methods to find application inputs that expose higher silent data corruption risk to conservatively evaluate reliability.
- **Input-Aware Adaptive Protection:** Developed adaptive lightweight protection techniques, improving reliability by protecting regions that are more likely to produce silent errors.
- **Data-Aware Efficient Decision Support:** Developed frameworks that help users efficiently choose compression decision based on storage needs, transfer constraints, and quality goals.

PUBLICATIONS

First-Author Peer-Reviewed Conference Publications

- **Deploying Lightweight Input-Aware Selective Instruction Duplication in HPC Applications**

Md Hasanur Rahman, Guanpeng Li

ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2025

Acceptance rate: 21.2%

- **Modeling Rate-Distortion for Endpoint-Aware Lossy Compression in Scientific Data Transfer**

Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello

IEEE International Performance Computing and Communications Conference (IPCCC), 2025

Acceptance rate: 32.1%

- **DRUTO: Upper-Bounding Silent Data Corruption Vulnerability in GPU Applications**

Md Hasanur Rahman, Sheng Di, Shengjian Guo, Xiaoyi Lu, Guanpeng Li, Franck Cappello

IEEE International Parallel & Distributed Processing Symposium (IPDPS), 2024

Acceptance rate: 25.0%

- **A Generic and Efficient Framework for Estimating Lossy Compressibility of Scientific Data**
Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
IEEE International Conference on Massive Storage Systems and Technology (MSST), 2024
- **A Feature-Driven Fixed-Ratio Lossy Compression Framework for Real-World Scientific Datasets**
Md Hasanur Rahman, Sheng Di, Kai Zhao, Robert Underwood, Guanpeng Li, Franck Cappello
IEEE International Conference on Data Engineering (ICDE), 2023
Acceptance rate: 19.1%
- **Peppa-X: Finding Program Test Inputs to Bound Silent Data Corruption Vulnerability in HPC Applications**
Md Hasanur Rahman, Aabid Shamji, Shengjian Guo, Guanpeng Li
ACM International Conference for High-Performance Computing, Networking, Storage and Analysis (SC), 2021
Acceptance rate: 23.6%

Journal and Survey Publications

- **A Survey on Error-Bounded Lossy Compression for Scientific Datasets**
Sheng Di, Jinyang Liu, Kai Zhao, Xin Liang, Robert Underwood, Zhaorui Zhang, Milan Shah, Yafan Huang, Jiajun Huang, Xiaodong Yu, Congrong Ren, Hanqi Guo, Grant Wilkins, Dingwen Tao, Jiannan Tian, Sian Jin, Zizhe Jian, Daoce Wang, Md Hasanur Rahman, Boyuan Zhang, Shihui Song, Jon Calhoun, Guanpeng Li, Kazutomo Yoshii, Khalid Alharthi, Franck Cappello
ACM Computing Surveys, 2025
- **Investigating the Impact of Transient Hardware Faults on Deep Learning Neural Network Inference**
Md Hasanur Rahman, Sabuj Laskar, Guanpeng Li
Journal of Software Testing, Verification and Reliability (STVR), 2024

Co-Authored Peer-Reviewed Conference Publications

- **Significantly Improving Fixed-Ratio Compression Framework for Resource-Limited Applications**
Tri Nguyen, Md Hasanur Rahman, Sheng Di, Michela Becchi
International Conference on Parallel Processing (ICPP), 2024
Acceptance rate: 29.0%
- **Characterizing Deep Learning Neural Network Failures between Algorithmic Inaccuracy and Transient Hardware Faults**
Sabuj Laskar, Md Hasanur Rahman, Bohan Zhang, Guanpeng Li
IEEE Pacific Rim International Symposium on Dependable Computing (PRDC), 2022
Acceptance rate: 36.0%

Workshop Publications

- **Characterizing Spatial Data Traits for Modeling Generic Lossy Rate-Distortion Quality**
Md Hasanur Rahman, Sheng Di, Guanpeng Li, Franck Cappello
IEEE International Parallel & Distributed Processing Symposium Workshops (IPDPS-W), 2025
- **LibPressio-Predict: Flexible and Fast Infrastructure for Inferring Compression Performance**
Robert R. Underwood, Sheng Di, Sian Jin, Md Hasanur Rahman, Arham Khan, Franck Cappello
ACM International Workshop on Data Reduction for Big Scientific Data (DRBSD), in conjunction with SC, 2023
- **TensorFI+: A Scalable Fault Injection Framework for Modern Deep Learning Neural Networks**

Sabuj Laskar, Md Hasanur Rahman, Guanpeng Li

IEEE International Workshop on Resiliency, Security, Defences and Attacks (ISSRE-W), 2022

INDUSTRY EXPERIENCE

Samsung Research

Software Engineer

2019–2021

Bangladesh

HONORS AND AWARDS

- SC Travel Grant, 2026.
- Graduate Research Excellence Award Finalist, 2025.
- SC Travel Grant, 2024.
- ICDE Travel Grant, 2023.

PROFESSIONAL SERVICE

- **Artifact Evaluation Committee:** IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2024.
- **Subreviewer:** DSN 2026; HiPC 2025; ISSRE 2024, 2023, 2022; HPDC 2023, 2022; DSN 2023, 2022; Middleware 2022; SELSE 2022; PRDC 2021.
- **Student Volunteer:** SC 2024; ICDE 2023.

STUDENT MENTORING

- Ahmer Jamil, 2025–2026.
- Abdullah Naveed, 2024–2026.
- Sabuj Laskar, 2022–2023.